

THEORY OF COMPUTATION (CS302)

10 Marks Question Bank

UNIT 1 – DFA & NFA

1. Explain DFA with a formal definition and construct a DFA for a given language.
2. Explain NFA and convert a given NFA into an equivalent DFA.
3. Differentiate DFA and NFA with suitable examples.
4. Design finite automata for specific pattern recognition problems.

UNIT 2 – Regular Languages & Pumping Lemma

1. Explain Regular Expressions and their applications.
2. Discuss Regular Languages with suitable examples.
3. State and explain the Pumping Lemma for Regular Languages.
4. Use Pumping Lemma to prove that a language is not regular.

UNIT 3 – Context-Free Grammars

1. Explain Context-Free Grammar and its components with examples.
2. Discuss Parse Trees and derivations in CFG.
3. Explain Ambiguous Grammars with examples.
4. Convert a given grammar into Chomsky Normal Form (CNF).

UNIT 4 – Pushdown Automata

1. Explain the structure and working of Pushdown Automata.
2. Design a PDA for the language $a^n b^n$.
3. Discuss acceptance by final state and empty stack.
4. Explain the equivalence between PDA and CFG.

UNIT 5 – Turing Machines & Decidability

1. Explain the structure and working of a Turing Machine.
2. Design a Turing Machine for a given language.
3. Discuss Decidable and Undecidable Problems.
4. Explain the Halting Problem and Church-Turing Thesis.

Exam Preparation Tips

Focus on automata construction problems, DFA/NFA conversions, Pumping Lemma proofs, CFG derivations, PDA designs, Turing Machine constructions, and Decidability theory as they frequently appear in university examinations.